Probabilistic Machine Learning

Lecture 06: Evaluation and Empirical Risk Minimisation

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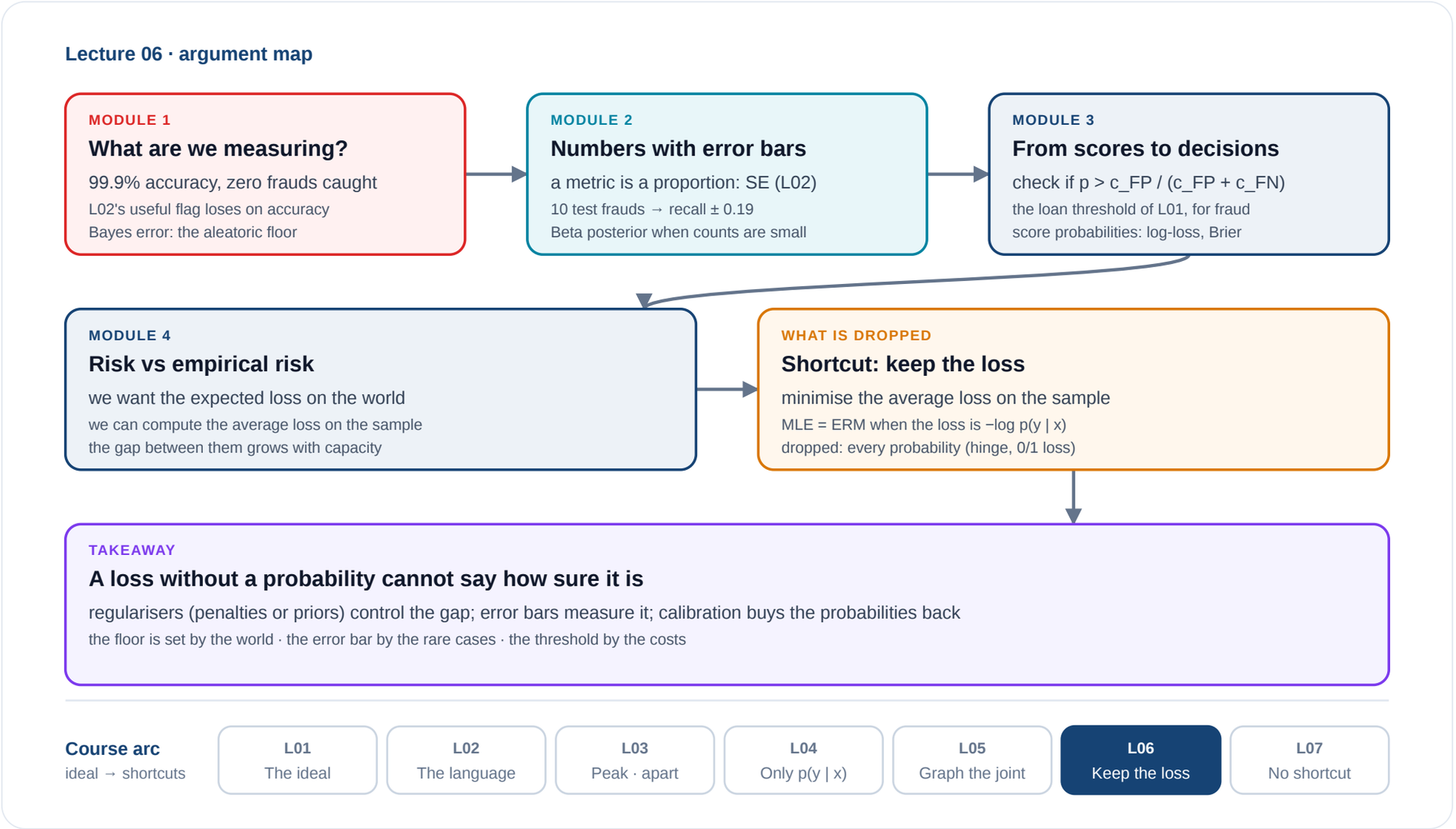
Speaker notes

Reworked 2026-09-26 (docs/reviews/PML-L05-L07-REVIEW-2026-09.md §5.2). The lecture evaluates probabilities, not only decisions, and it names its shortcut. It stays on Lecture 02's fraud model throughout: Module 1 shows that the useful flag loses to "always legitimate" on accuracy and that the Bayes error is the aleatoric floor of Lecture 01; Module 2 puts error bars on metrics (Lecture 02's SE and Beta posterior); Module 3 turns calibrated scores into decisions with Lecture 01's cost threshold and scores the probabilities themselves; Module 4 names what training minimised all along — the empirical risk — and what that shortcut drops.

Practice first, theory last: the students do evaluation before they are told that training was ERM, so the theory arrives as an explanation. The Data Science evaluation units (measuring-performance-*) are no longer included; CV schemes and precision/recall are recapped in one slide each, because the students have taken ML courses before.



Argument Map



Speaker notes

Read it top-left to bottom. Module 1: on fraud, accuracy rewards the model that catches nothing, and even the best model has a floor (the Bayes error). Module 2: every metric is an estimate, and for a rare class its error bar is set by the rare cases. Module 3: a calibrated probability plus the costs gives the decision, and the probabilities themselves can be scored. Module 4 names the shortcut in the strip: training minimised an average loss on the sample; unless that loss was a negative log-probability, the probabilities were dropped — which is why Modules 2 and 3 had to buy them back.

Module 1

What Are We Measuring?

Guiding Question: A fraud model reports 99.9% accuracy and has never caught a single fraud. What should we have measured instead — and how low could the error of the best model be?

What We Want to Know

The **true error** of a classifier h is the probability that it errs on a fresh example drawn from the world:

$$\mathrm{err}^*(h) = P\big(h(X) \neq Y\big), \qquad (X, Y) \sim p$$

The **Bayes error** is the smallest true error any classifier can reach:

$$\min_h \mathrm{err}^*(h) = \mathbb{E}_X\Big[\min\big(p(y{=}1 \mid X),\ \ 1 - p(y{=}1 \mid X)\big)\Big]$$

A transaction type is fraudulent 55% of the time. Could a better model push its error below 45%?

Speaker notes

Math first (P5): let the room read the second formula before interpreting it. The best rule predicts the more probable label for every x, and on each x it is wrong with the probability of the less probable one; averaging over x gives the formula. Y is a random variable, not a function of x — that is the whole point. The 0.55 is Lecture 01's opening score. Push the nuance from Lecture 01 again: the floor is relative to the attributes; a new attribute (the device, the location) can lower it, more rows of the same attributes cannot.

What we can compute is the error on a sample — the rest of the lecture is about how much that number can be trusted, and whether it was the right number.

What We Want to Know

The **true error** of a classifier h is the probability that it errs on a fresh example drawn from the world:

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A transaction type is fraudulent 55% of the time. Could a better model push its error below 45%?

No — only **new attributes** can. The Bayes error is aleatoric uncertainty (Lecture 01's identical patients) written as an error rate: the floor that no amount of data removes.

Lecture 02's Flag, Evaluated

per 100,000 transactions, 100 frauds	always "legitimate"	Lecture 02's flag
frauds caught · honest customers flagged	0 · 0	99 · 999
accuracy	**99.9%**	**99.0%**
recall $P(\text{flag} \mid \text{fraud})$	0%	99%
precision $P(\text{fraud} \mid \text{flag})$	—	9%

Which model would you deploy — and which one does accuracy recommend?

Speaker notes

This is the payoff of Lecture 02's promise ("Lecture 06 opens with exactly that model"). The numbers are the same table: 99 + 98,901 correct gives 99.0%, and 99,900 correct gives 99.9%. Let the room vote before the click. Then name the two rows below accuracy as what students already know as recall and precision — and point out that Lecture 02 computed them as two conditionals of Bayes' theorem: P(flag | fraud) = 0.99 and P(fraud | flag) = 99/1,098.

Lecture 02's Flag, Evaluated

per 100,000 transactions, 100 frauds	always "legitimate"	Lecture 02's flag
frauds caught · honest customers flagged	0 · 0	99 · 999
accuracy	**99.9%**	**99.0%**
recall $P(\text{flag} \mid \text{fraud})$	0%	99%
precision $P(\text{fraud} \mid \text{flag})$	—	9%

Which model would you deploy — and which one does accuracy recommend?

> By accuracy, the model that catches **nothing** wins. On a rare class, accuracy counts the majority and hides that a model is useless.

What to Report Instead

- → **A baseline** next to every number: always-majority for classification, the mean for regression. "99.9%" is not a result until you know that doing nothing scores 99.9%.
- → **Measures that condition on the rare class**: recall conditions on the truth, precision on the decision — the two directions of Bayes' theorem that Lecture 02 warned not to swap.
- → **The costs**: which of the two mistakes the application pays for. Module 3 turns them into a threshold.

> Precision depends on the base rate, recall does not: the same flag has precision 9% at 1 in 1,000 and about 50% at 1 in 100.



Speaker notes

A list, not cards: these are three habits, not three contrasting positions. The last line checks whether Lecture 02 landed: at prevalence 1/100, the flag catches 990 of 1,000 frauds and flags 990 of 99,000 honest customers, so precision is 990/1,980 = 0.5. Recall stays 0.99. That is why precision is the number that surprises people when a model moves to a new market.

Mini-Exercise

A model flags **every** transaction as fraud (one in 1,000 is fraudulent).

What are its recall and its precision?



Speaker notes

One minute to read, one to think. The mirror image of the always-legitimate model: that one maximises accuracy, this one maximises recall. Neither number alone says anything about usefulness.

Mini-Exercise

A model flags **every** transaction as fraud (one in 1,000 is fraudulent).

What are its recall and its precision?

Recall **100%** — it catches every fraud. Precision **0.1%** — the base rate. (Accuracy: 0.1%.)
Recall alone can be bought by flagging everything; that is why it is always reported together with precision, or with the costs.



Takeaway

We want the **true error**, whose floor — the **Bayes error** — is aleatoric uncertainty that no data removes. On a rare class, accuracy measures the majority: report a **baseline** and measures that condition on the rare class.

Module 2

Numbers With Error Bars

Guiding Question: The test set says recall 0.90. Would you promise the bank that the model catches 90% of next month's frauds?

Where an Error Number Comes From

estimate	how	what it is
training error	evaluate on the data the model was fitted on	optimistically biased — on average below the true error of the fitted model, often far below
hold-out	split once; fit on one part, evaluate on the other	an unbiased estimate of err^* — for **one** split, with a sampling error
k-fold cross-validation	each of k folds serves once as the test set; average	uses every example for testing once; still an estimate from a sample

> A test set that is consulted for choices — which model, which threshold — becomes training data. Keep one set that is looked at once.

Speaker notes

Recap for ML-experienced students: one slide, the three schemes they know, with the two statements that are often said wrongly. Training error is not a *lower bound* on the true error (for a given sample it can be above it); it is biased in expectation for the model it selected. Hold-out is one split, not "2-fold". The warning is the leaderboard effect. The rest of the module asks the question none of the three rows answers: how precise is the number?

A Metric Is a Proportion

Accuracy on n test cases is the mean of n yes/no indicators — a Bernoulli sample (Lecture 02). Its standard error is $\sigma/\sqrt{n}$ with $\sigma^2 = p(1-p)$:

$$\mathrm{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

A test set of **10,000** transactions, one in 1,000 fraudulent:

0.990 ± 0.002

accuracy: $n = 10{,}000$, $\mathrm{SE} \approx 0.001$

0.90 ± 0.19

recall, 9 of 10 frauds caught: $n = 10$, $\mathrm{SE} \approx 0.095$

Same test set, same model — why is one error bar a hundred times wider?

Speaker notes

Both numbers are ±1.96 SE, rounded. SE(accuracy) = √(0.99 · 0.01 / 10,000) ≈ 0.001; SE(recall) = √(0.9 · 0.1 / 10) ≈ 0.095, so the 95% interval is [0.71, 1.09] — it runs past 1. That is the normal approximation breaking at n = 10 (Lecture 02 noted the same for 3 of 10 users) and the reason for the next slide.

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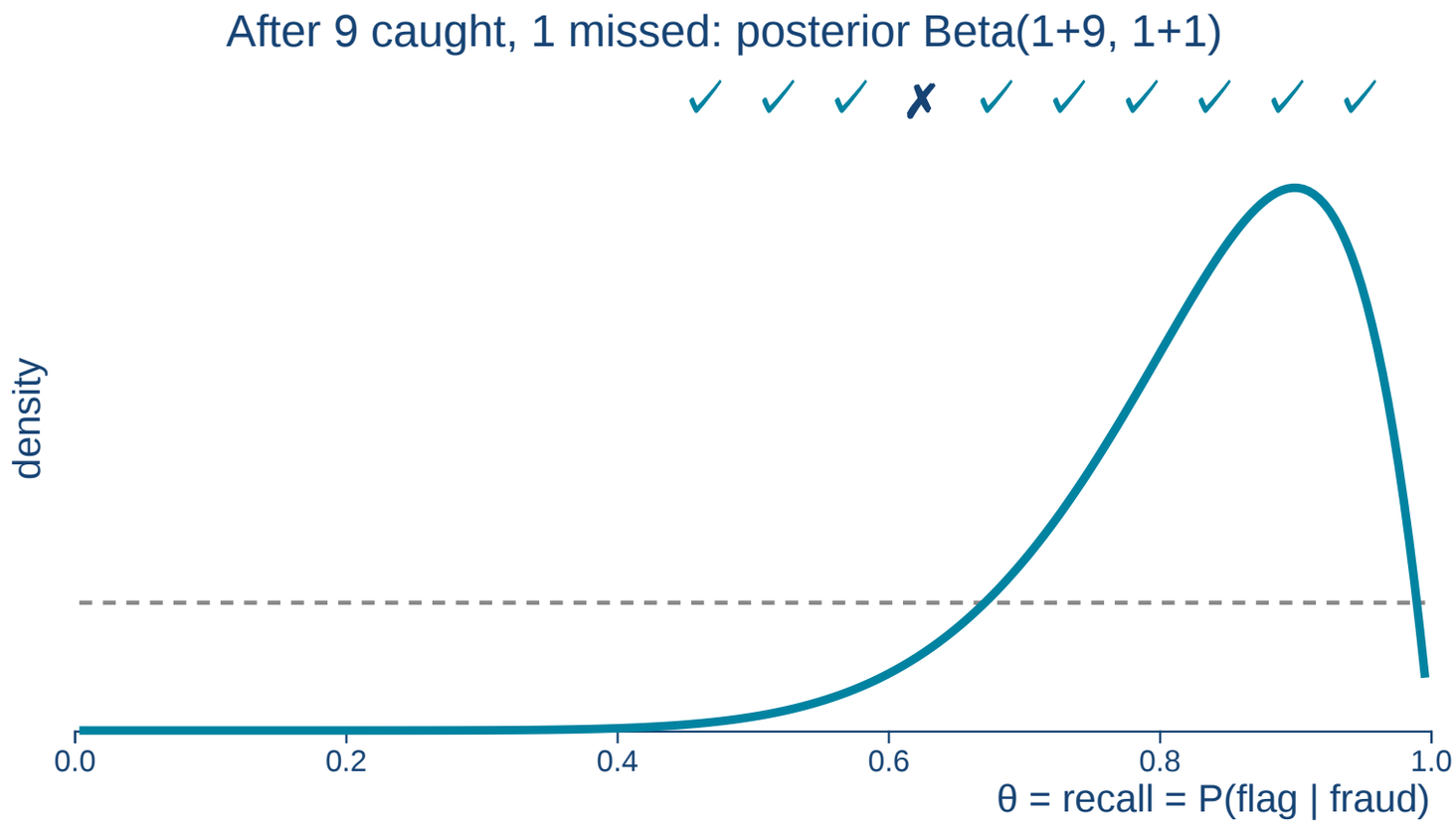
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recall, 9 of 10 frauds caught: $n = 10$, $\mathrm{SE} \approx 0.095$

Same test set, same model — why is one error bar a hundred times wider?

Recall is a proportion **of the frauds**, and there are only ten of them. The error bar is set by the rare class, not by the size of the test set.

Recall as an Unknown Parameter



After 9 caught and 1 missed, with a flat prior: $\mathrm{Beta}(10, 2)$ — mean **0.83**, 95% credible interval **[0.59, 0.98]**.

Speaker notes

The Beta–Bernoulli model of Lecture 02 with "user returns" replaced by "fraud caught": each test fraud is one observation of the recall. Click through; the curve narrows slowly because there are only ten observations. The credible interval stays inside [0, 1] by construction and says what everyone wanted to say: the recall is probably somewhere between 0.6 and 0.98. The posterior mean 0.83 sits below the observed 0.9 because the flat prior counts one catch and one miss (Lecture 02's pseudo-counts). With 100 test frauds, 90 caught, the interval would be [0.83, 0.94].

The Rare Class Sets the Test Size

Recall near 0.9, so $\mathrm{SE} = \sqrt{0.09/n_{\text{fraud}}}$; at one fraud in 1,000 transactions:

frauds in the test set	transactions	SE of recall	error bar (±2 SE)
10	10,000	0.095	±0.19
100	100,000	0.030	±0.06
400	400,000	0.015	±0.03
1,000	1,000,000	0.0095	±0.02

Model B catches 10 of the 10 test frauds, model A catches 9. That is **one fraud** of difference — well inside either error bar. Compare two models on the **same** test cases and look at the cases where they disagree.

Speaker notes

The 1/√n law of Lecture 02 ("precision costs data"), now with a price tag in transactions: halving the error bar needs four times the frauds, and every fraud costs a thousand transactions. This is why fraud teams evaluate on months of data. The paired comparison is the honest way to compare two models: cases both get right or both get wrong carry no information about which is better; only the disagreements do. If A and B disagree on a single fraud, there is nothing to conclude.

Mini-Exercise

Recall is near 0.9, so $\mathrm{SE} = \sqrt{0.09/n}$ for n test frauds.

How many frauds does the test set need for $\mathrm{SE} = 0.03$ — and how many transactions is that?

Speaker notes

Head arithmetic: $0.03^2 = 0.0009$. The second sentence is the transfer: the square-root law again. Students who reach for 1.96 have understood the next step; the question is only about the SE.

Mini-Exercise

Recall is near 0.9, so $\mathrm{SE} = \sqrt{0.09/n}$ for n test frauds.

How many frauds does the test set need for $\mathrm{SE} = 0.03$ — and how many transactions is that?

$0.09/n = 0.0009$, so $n = 100$ frauds — about **100,000 transactions** at one in 1,000. For $\mathrm{SE} = 0.015$ it would be four times as many.

Takeaway

> Every metric is an **estimate from a sample** and needs an **error bar**: a standard error, or a Beta posterior when the counts are small. For a rare class the error bar is set by the number of **rare cases**, not by the size of the test set.

Module 3

From Scores to Decisions

Guiding Question: The model says p(fraud) = 0.09. Check the transaction or let it through — and who decides where the line is?

Two Mistakes, Two Prices

	transaction is fraud	transaction is legitimate
check it (a text to the customer)	0	€5 (c_{FP})
let it through	€500 (c_{FN})	0

The model says $p = p(\text{fraud} \mid \mathbf{x})$. Expected cost of checking: $(1 - p) \cdot c_{FP}$; of letting it through: $p \cdot c_{FN}$.

At which p does checking become the cheaper action?

Speaker notes

Math first, then the question. The derivation is one line: check iff (1 − p)·c_FP < p·c_FN. Students have seen it in Lecture 01 (€1,000 against €9,000, p* = 0.1). The 9% precision that looked damning in Lecture 02 now looks different: precision is not the criterion; expected cost is. Correct outcomes cost 0 here; a checked fraud could cost €5 as well, which changes p* only marginally — say so if asked.

Two Mistakes, Two Prices

	transaction is fraud	transaction is legitimate
check it (a text to the customer)	0	€5 (c_{FP})
let it through	€500 (c_{FN})	0

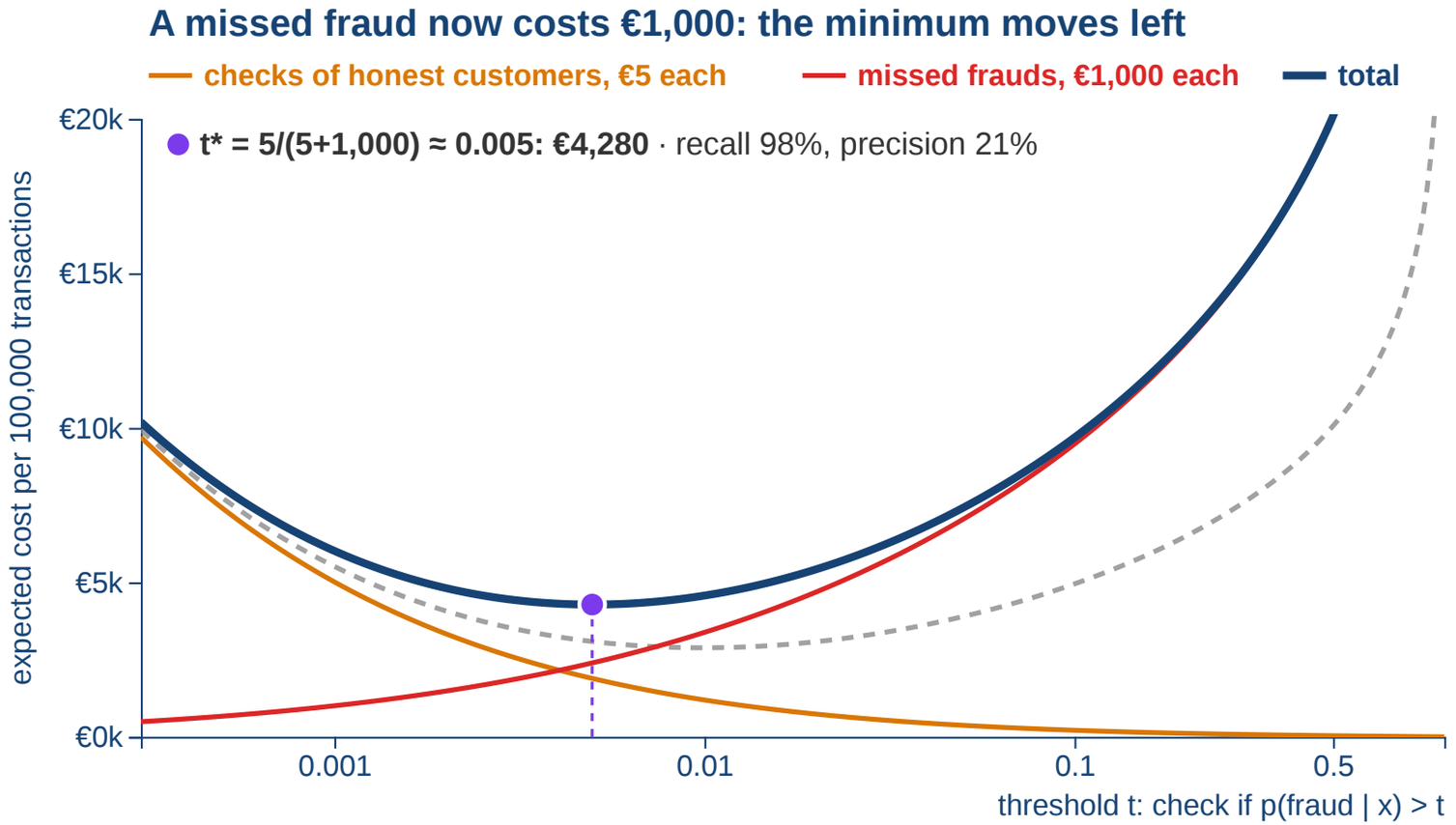
The model says $p = p(\text{fraud} \mid \mathbf{x})$. Expected cost of checking: $(1 - p) \cdot c_{FP}$; of letting it through: $p \cdot c_{FN}$.

At which p does checking become the cheaper action?

$$p > p^* = \frac{c_{FP}}{c_{FP} + c_{FN}} = \frac{5}{505} \approx 0.01$$

> Lecture 01's loan threshold, for fraud. Lecture 02's flagged transaction (9% fraud) is worth checking: 9% is nine times the break-even.

Sweeping the Threshold



Predict before the last click: a missed fraud now costs €1,000. Does the best threshold move left or right — and does recall go up or down?

Speaker notes

Stepped viz (P6). A calibrated model on 100,000 transactions, one in 1,000 fraudulent. On entry: the two costs — checking honest customers falls as the threshold rises, missed frauds rise. Click 1: their sum. Click 2: the minimum, at t* ≈ 0.01, exactly c_FP/(c_FP + c_FN) (€2,881: recall 97%, precision 29%). Click 3: Lecture 02's flag is this model at t = 0.001 (recall 99%, precision 9%, €5,515); the default t = 0.5 catches only 80% and costs €10,062 — three and a half times the optimum. Click 4: c_FN = €1,000 moves t* to 0.005 (recall 98%, €4,280). Say it out loud: precision and recall are not properties of a model but of a model plus a threshold, and the costs choose the point. The threshold rule works only if the score is **calibrated** — Lecture 04's condition.

Scoring the Probabilities Themselves

A threshold uses the score once. To evaluate the **probabilities**, score every prediction on held-out data:

Log-loss

$$\text{log-loss} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{\text{model}}(y_i \mid \mathbf{x}_i)$$

the held-out negative log-likelihood — Lecture 04's cross-entropy, on test data

Brier score

$$\text{Brier} = \frac{1}{n}\sum_{i=1}^{n} \left(p_i - y_i\right)^2$$

squared error of the probability against the 0/1 outcome

Both scores are **proper**: in expectation they are best when the model reports the true probability. Accuracy is not — it cannot tell 0.51 from 0.99.

Speaker notes

Log-loss is the one the course has used all along: maximum likelihood (Lecture 03) minimised it on the training data; here it is measured on data the model has not seen. "Proper" is the property that matters: a model cannot improve its expected score by reporting anything other than what it believes — which is exactly the incentive a decision-maker needs from a probability.

What the Scores Punish

the model said $p = 0.09$, and the transaction was…	log-loss term	Brier term
fraud	2.41	0.83
legitimate	0.09	0.008
fraud, but the model had said $p = 0.001$	**6.91**	0.998

Log-loss punishes **confident errors** without limit — $p = 0$ on a fraud costs infinity; Brier is bounded by 1. Whether "0.09" comes true 9% of the time is Lecture 04's **reliability diagram**, now drawn on held-out data.

Speaker notes

The numbers: $-\ln 0.09 = 2.41$, $-\ln 0.91 = 0.094$; $(0.91)^2 = 0.83$, $(0.09)^2 = 0.008$; $-\ln 0.001 = 6.91$, $(0.999)^2 = 0.998$. That is also why the always-legitimate model, right 99.9% of the time, is exposed at once when it states its certainty as a probability: one fraud at $p = 0$ and its log-loss is infinite. Refer to the reliability diagram of Lecture 04 rather than redrawing it: the new point is only that it must be drawn on data the model has not seen.

Mini-Exercise

Checking costs €5, a missed fraud €500. The model says $p = 0.02$.

Check the transaction or let it through?

Speaker notes

Two minutes, head arithmetic. A 2% fraud probability sounds negligible; at a 100:1 cost ratio it is twice the break-even. The same reversal as the loan exercise of Lecture 01.

Mini-Exercise

Checking costs €5, a missed fraud €500. The model says $p = 0.02$.

Check the transaction or let it through?

Check. Letting it through costs 0.02 · €500 = €10 in expectation; checking costs 0.98 · €5 ≈ €4.90. Equivalently, $0.02 > p^* \approx 0.01$.

Takeaway

A **calibrated probability** plus the **costs** gives the decision: act when $p > c_{FP}/(c_{FP} + c_{FN})$. Precision and recall move along one curve as the threshold moves; **log-loss** and the **Brier score** evaluate the probabilities themselves.

Module 4

Keep the Loss: Empirical Risk Minimisation

Guiding Question: Every model you trained minimised something on the training data. What did we actually want — and what did the substitution throw away?

What We Actually Want to Minimise

The **risk** of a hypothesis h under a loss L is its expected loss on the world:

$$R(h) = \mathbb{E}_{(\mathbf{x},y)\sim p}\big[L\big(h(\mathbf{x}), y\big)\big]$$

0/1 loss

$R(h) = \mathrm{err}^*(h)$ — the true error of Module 1

Cost matrix

$R(h)$ = the expected cost of Module 3

It is the quantity that matters — and **uncomputable**: p is unknown.

Speaker notes

Math first (P5). Everything measured in Modules 1–3 was a risk under some loss: the error rate under 0/1 loss, the expected cost under the fraud cost matrix. The gap between what we want (an expectation under p) and what we can compute (an average over a sample) is the whole module.

What Training Computes Instead

The **empirical risk** is the average loss on the training sample, and **empirical risk minimisation (ERM)** picks its minimiser:

$$\hat{R}(h) = \frac{1}{n}\sum_{i=1}^{n} L\big(h(\mathbf{x}_i), y_i\big), \qquad \hat{h} = \arg\min_{h \in \mathcal{H}} \hat{R}(h)$$

loss	trained model	is $h(\mathbf{x})$ a probability?
squared error	linear regression	the mean of a Gaussian — **yes** (Lecture 04)
cross-entropy $-\log p(y \mid \mathbf{x})$	logistic regression, most neural nets	**yes** — this is maximum likelihood for $p(y \mid \mathbf{x})$
hinge $\max(0, 1 - y \cdot h(\mathbf{x}))$	support vector machine	**no** — a distance to the margin
0/1	what we care about; no useful gradient	**no** — only a label

Speaker notes

A table, because the rows are compared on shared dimensions. The last column is the one the lecture needs: ERM is a general recipe, and only some losses keep a probability behind the number. Correct the tempting overstatement explicitly: "nearly every method is ERM" is false in this course. The Beta posterior (Lecture 02), inference in a Bayesian network (Lecture 05) and Bayesian regression (Lecture 07) condition on data; nothing is minimised.

Keep the Loss, Drop the Probabilities

WHAT ERM WITH A HINGE OR 0/1 LOSS RETURNS

a score: 2.3

large means "more fraud-like" — nothing more

WHAT IT CAN NO LONGER DO

- apply the cost threshold $p > c_{FP}/(c_{FP} + c_{FN})$ of Module 3
- be scored by log-loss or Brier, or drawn in a reliability diagram
- put an error bar on its own confidence

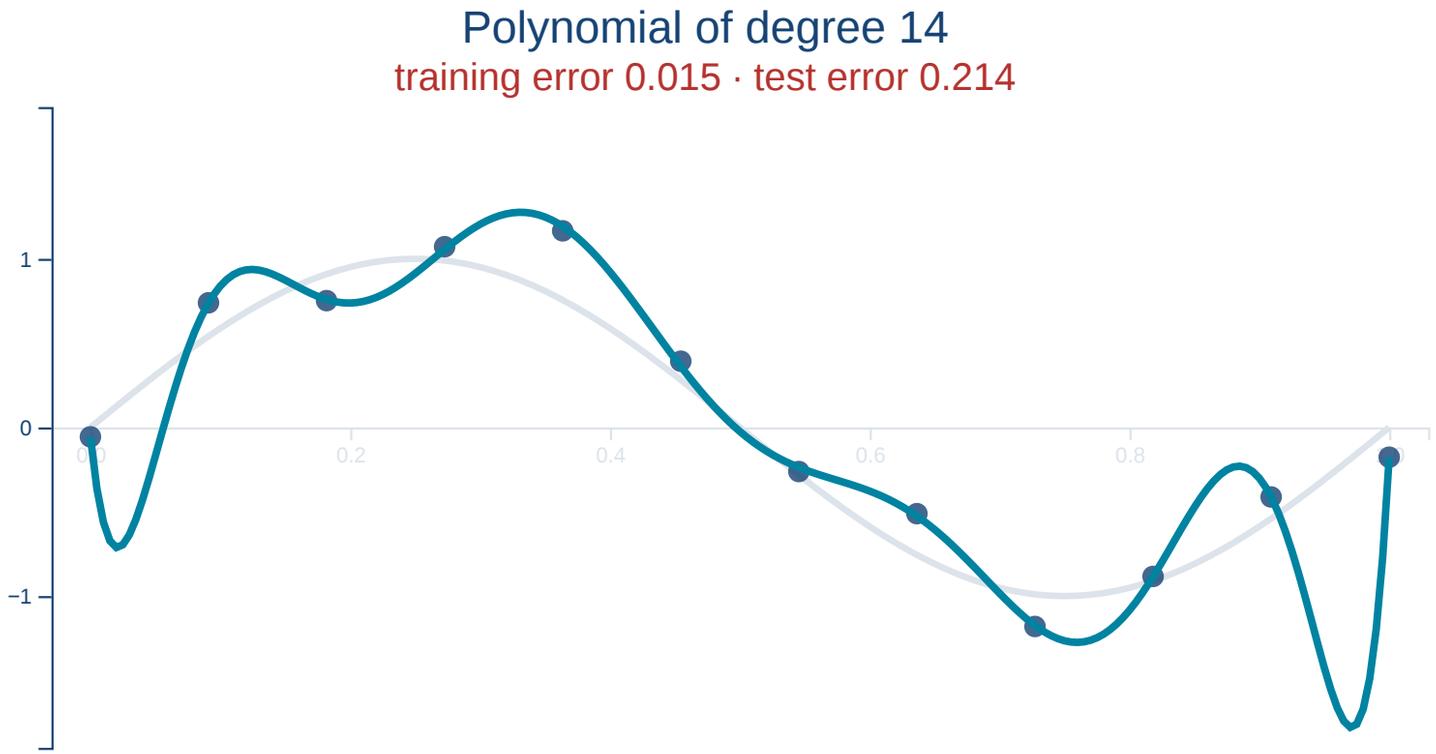
To use it for a decision, **buy the probabilities back**: fit $p = \sigma(a \cdot \text{score} + b)$ on held-out data (Platt scaling) — a logistic regression on the score, i.e. calibration.

Not every method is ERM: maximum likelihood for $p(y \mid \mathbf{x})$ is ERM with loss $-\log p$; generative maximum likelihood (naive Bayes) scores $\log p(\mathbf{x}, y)$, and Bayesian methods minimise nothing.

Speaker notes

The slide that links Modules 3 and 4. Ask first: "the SVM says 2.3 — check the transaction?" Nobody can answer without knowing what 2.3 means in probability, which is exactly what was dropped. Platt scaling is one line of Lecture 04: logistic regression with the score as the only feature, fitted on data the SVM did not train on. Even cross-entropy models are often miscalibrated after heavy training, so the reliability check of Module 3 applies to them too.

The Generalisation Gap



$R(\hat{h}) = \hat{R}(\hat{h}) + \big(R(\hat{h}) - \hat{R}(\hat{h})\big)$: the gap opens because $\hat{h}$ was **chosen on the data that then measures it**, and it tends to grow with the capacity of $\mathcal{H}$.

Speaker notes

Stepped viz: each click raises the polynomial degree on the same twelve training points. Training error falls every time; test error falls, bottoms out, then climbs. Two precise statements to add. (1) Training error is an unbiased estimate of R(h) for a fixed h chosen without that data — it is biased only for the h it selected. (2) A richer H widens the gap but can lower the risk: the excess risk over the best possible splits into approximation error (H too poor — falls with capacity) and estimation error (too little data to find the best h in H — grows with capacity). That split, not the gap, is the risk version of bias–variance.

Structural Risk Minimisation — and the Prior Inside It

Minimise the empirical risk **plus** a complexity penalty:

$$\hat{h} = \arg\min_{h \in \mathcal{H}} \hat{R}(h) + \lambda \Omega(h)$$

With the loss $-\log p(y \mid \mathbf{x}, \mathbf{w})$, the MAP estimate of Lecture 04 has exactly this form:

$$\frac{1}{n} \sum_{i=1}^{n} -\log p(y_i \mid \mathbf{x}_i, \mathbf{w}) \quad \underbrace{-\frac{1}{n} \log p(\mathbf{w})}_{\lambda \Omega(\mathbf{w})}$$

> The $\frac{1}{n}$ is the point: the prior's weight **fades as data grows** — the pseudo-counts of Lecture 02's Beta prior. A Gaussian prior $\mathcal{N}(0, \tau^2 \mathbf{I})$ gives $\lambda \Omega(\mathbf{w}) = \|\mathbf{w}\|^2 / (2n\tau^2)$.

Speaker notes

Divide the negative log-posterior by n to put it on the scale of an average loss; the prior term picks up the 1/n. Penalty, capacity control and prior are three readings of one term. But note what even MAP keeps: a single peak. The regulariser controls the gap; it does not bring back how sure the model is — that needs the whole posterior (Lecture 07).

Mini-Exercise

A linear SVM gives a transaction the score **2.3**. The fraud team checks every transaction with $p > 0.01$.

Can they apply their rule to this transaction?

Speaker notes

One question, the module's one point. Some students will answer "2.3 > 0.01, check": that is the confusion the slide is for. Others will say "the score is positive, so fraud": that is the SVM's own 0/1 decision, which uses the threshold 0 — and assumes equal costs.

Mini-Exercise

A linear SVM gives a transaction the score **2.3**. The fraud team checks every transaction with $p > 0.01$.

Can they apply their rule to this transaction?

Not directly. 2.3 is a distance to the margin, not a probability — the hinge loss dropped it. First calibrate on held-out data, $p = \sigma(a \cdot 2.3 + b)$, then compare that p with 0.01.

Takeaway

Training minimises the **empirical risk** because the **risk** is out of reach. ERM keeps the **loss** and — unless the loss is a negative log-probability — drops the **probabilities**. Regularisers (penalties or priors) control the gap; error bars measure it; calibration buys the probabilities back.